

GavinKlorfine–Assign4

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2026-02-04

DDAR Exercise 5.1

The data set criminal in the package logmult gives the 4×5 table below of the number of men aged 15–19 charged with a criminal case for whom charges were dropped in Denmark from 1955–1958.

Part (a)

Use loglm() to test whether there is an association between Year and Age. Is there evidence that dropping of charges in relation to age changed over the years recorded here?

```
data("criminal", package = "logmult")

(mod_ageYr <- loglm(~Age + Year, data = criminal))
```

```
## Call:
## loglm(formula = ~Age + Year, data = criminal)
##
## Statistics:
##              X^2 df      P(> X^2)
## Likelihood Ratio 38.24466 12 0.0001400372
## Pearson          38.41033 12 0.0001315495
```

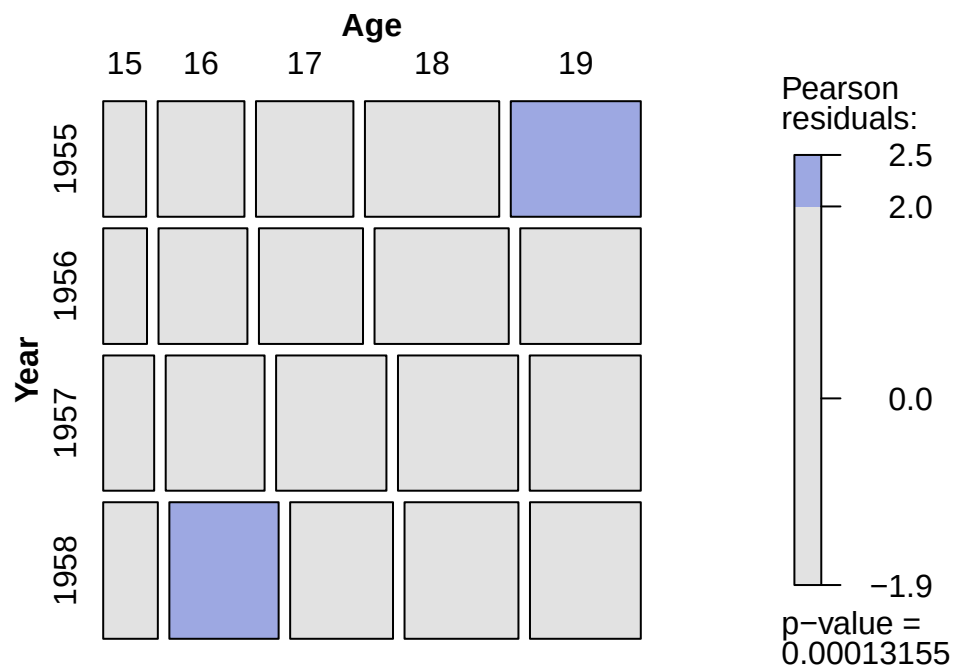
Due to a significant departure from independence, $\chi^2(12) = 38.41$, $p < .001$, there is evidence that dropping of charges in relation to age changed over the years recorded.

Part (b)

Use mosaic() with the option shade=TRUE to display the pattern of signs and magnitudes of the residuals. Compare this with the result of mosaic() using “Friendly shading,” from the option gp=shading_Friendly. Describe verbally what you see in each regarding the pattern of association in this table.

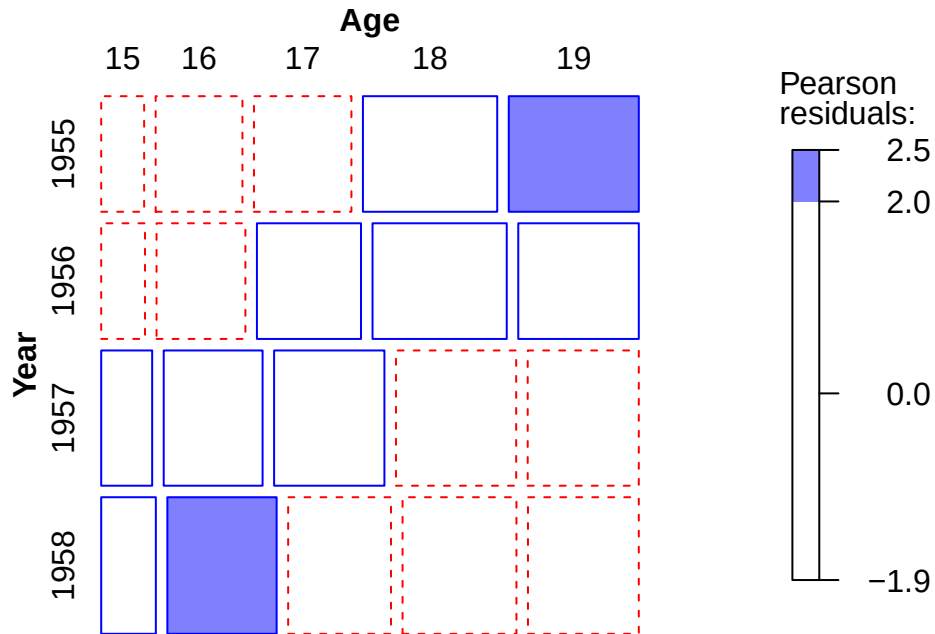
```
def_shade <- mosaic(criminal, shade = TRUE,
                    main = "Criminal Data, Default Shading")
```

Criminal Data, Default Shading



```
mf_shade <- mosaic(criminal, shade = TRUE, gp = shading_Friendly,  
  main = "Criminal Data, Friendly Shading")
```

Criminal Data, Friendly Shading



Since most of the residuals lie within the interval ± 2 , the default shading scheme does not readily depict a clear pattern regarding cells' relative contributions to the overall χ^2 statistic. In contrast, the mosaic plot with the Friendly shading scheme reveals the sign of cells within the interval ± 2 , and consequently, a diagonal pattern of positive residuals. The interpretation here may be that as year (1955-1958) increases, men of younger age (from the interval 15-19) have larger than expected (under independence) frequencies of charges being dropped.

DDAR Exercise 5.4

A related analysis concerns differences in throwing handedness among baseball players according to the fielding position they play. The following code calculates such a frequency table.

```
library(Lahman)
# NOTE: It seems the "Master" data is now the "People" data
data("People", package = "Lahman")
MasterFielding <- data.frame(merge(People, Fielding, by= "playerID"))
throwPOS <- with(MasterFielding, table(POS, throws))
```

Part (a)

Make a mosaic display of throwing hand vs. fielding position.

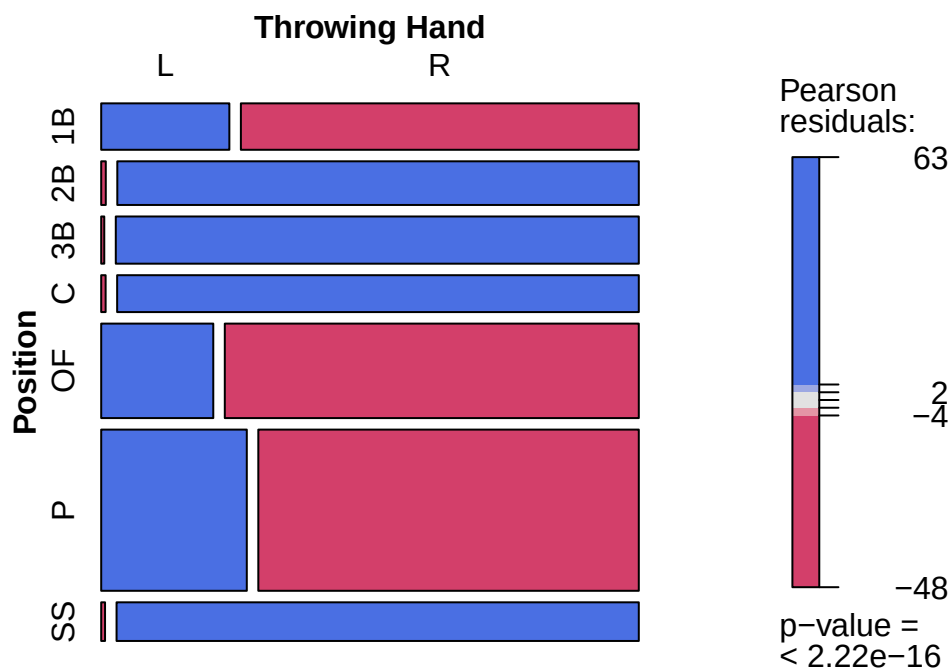
```
head(throwPOS)
```

```
##      throws
## POS      B      L      R      S
## 1B      9 3650 11332      0
## 2B      5  123 13962      0
## 3B      6   89 15146      0
## C       0  100 11802      0
## OF     10 6497 23996      0
## P      21 14398 37623      6
```

```
throwPOS <- throwPOS[,2:3] # Remove columns B/S
names(dimnames(throwPOS)) <- c("Position", "Throwing Hand")

mosaic(throwPOS, shade = TRUE,
       main = "Throwing hand Versus Fielding Position\nFrom the People Dataset")
```

Throwing hand Versus Fielding Position From the People Dataset



Part (b)

Calculate the percentage of players throwing left-handed by position. Make a sensible graph of this data.

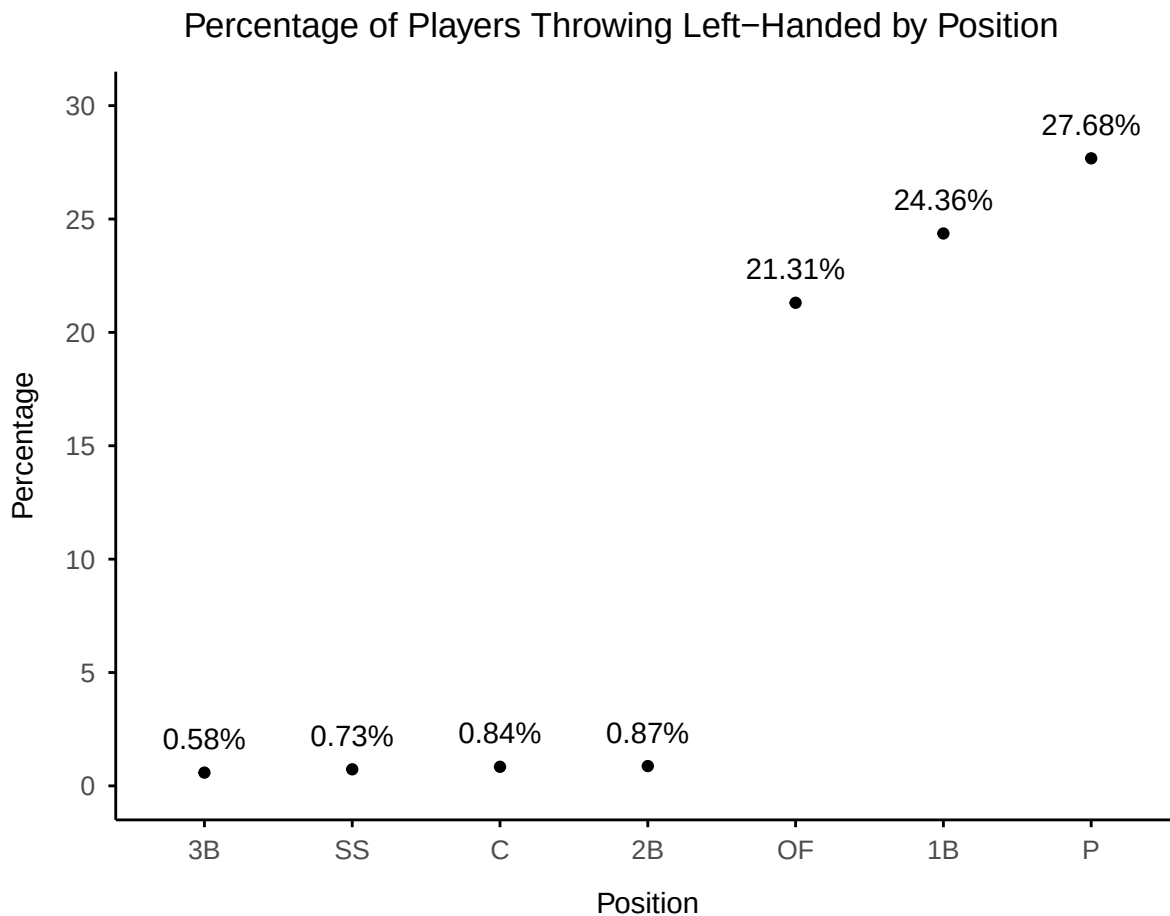
```
(pc_left <- 100 * throwPOS[,1] / (throwPOS[,1] + throwPOS[,2]))
```

```
##           1B           2B           3B           C           OF           P           SS
## 24.3625684  0.8732694  0.5841812  0.8401949 21.3065294 27.6772842  0.7299270
```

```
pc_left_df <- data.frame(
  Position = names(pc_left),
  throw_left = throwPOS[,1],
  throw_right = throwPOS[,2],
  Percentage_Left = unname(pc_left),
  row.names = NULL
)

pc_left_df <- pc_left_df |> arrange(Percentage_Left)
pc_left_df$Position <- factor(pc_left_df$Position, levels = pc_left_df$Position)

ggplot(pc_left_df, aes(x = Position, y = Percentage_Left)) +
  geom_point() +
  geom_text(
    label = paste0(round(pc_left_df$Percentage_Left, 2), "%"),
    nudge_y = 1.5
  ) +
  coord_cartesian(ylim = c(0, 30)) +
  scale_y_continuous(breaks = seq(0, 30, 5)) +
  labs(
    title = "Percentage of Players Throwing Left-Handed by Position",
    y = "Percentage"
  ) +
  gavinTheme()
```



Part (c)

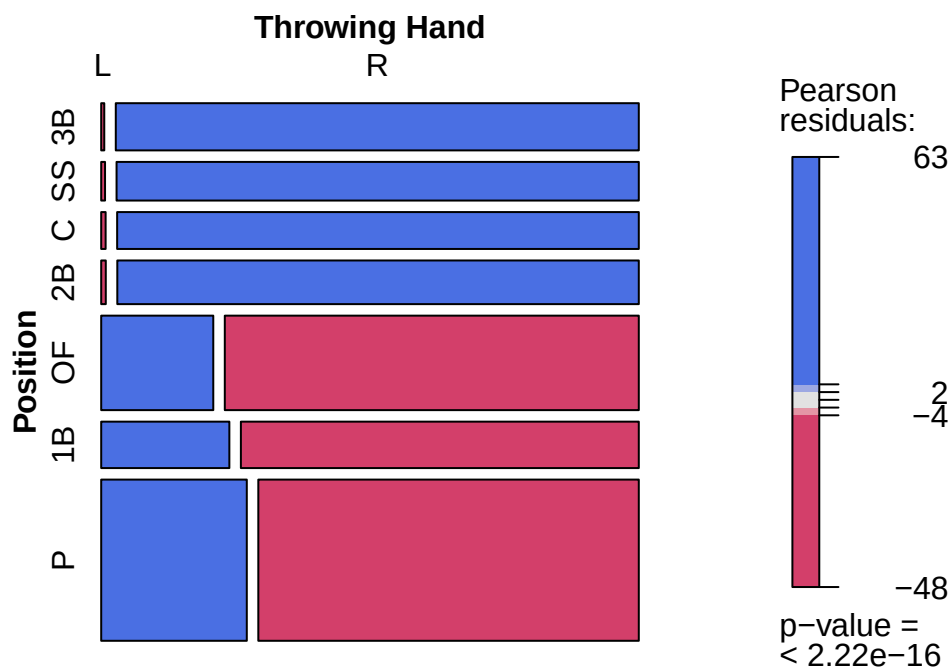
Re-do the mosaic display with the positions sorted by percentage of left-handers.

```
throwPOS_sort <- cbind( # Create table based on sorted dataframe
  xtabs(throw_left~Position, pc_left_df),
  xtabs(throw_right~Position, pc_left_df)
)

colnames(throwPOS_sort) <- c("L", "R")
names(dimnames(throwPOS_sort)) <- c("Position", "Throwing Hand")

mosaic(throwPOS_sort, shade=TRUE,
  main = "Throwing hand Versus Fielding Position\nFrom the People Dataset")
```

Throwing hand Versus Fielding Position From the People Dataset



Part (d)

Is there anything you can say about positions that have very few left-handed players?

Other than 1B, all positions with few left-handed players are in the infield as a consequence of most plays requiring a throw to the infielder's left-hand side (to 1B). The first-baseman does not have this issue as throws would most often be towards their right-hand side (i.e., to 2B and 3B but not home).

DDAR Exercise 5.8

The HairEyePlace data in vcdExtra gives similar data on hair color and eye color, for both Caithness and Aberdeen as a $4 \times 5 \times 2$ table.

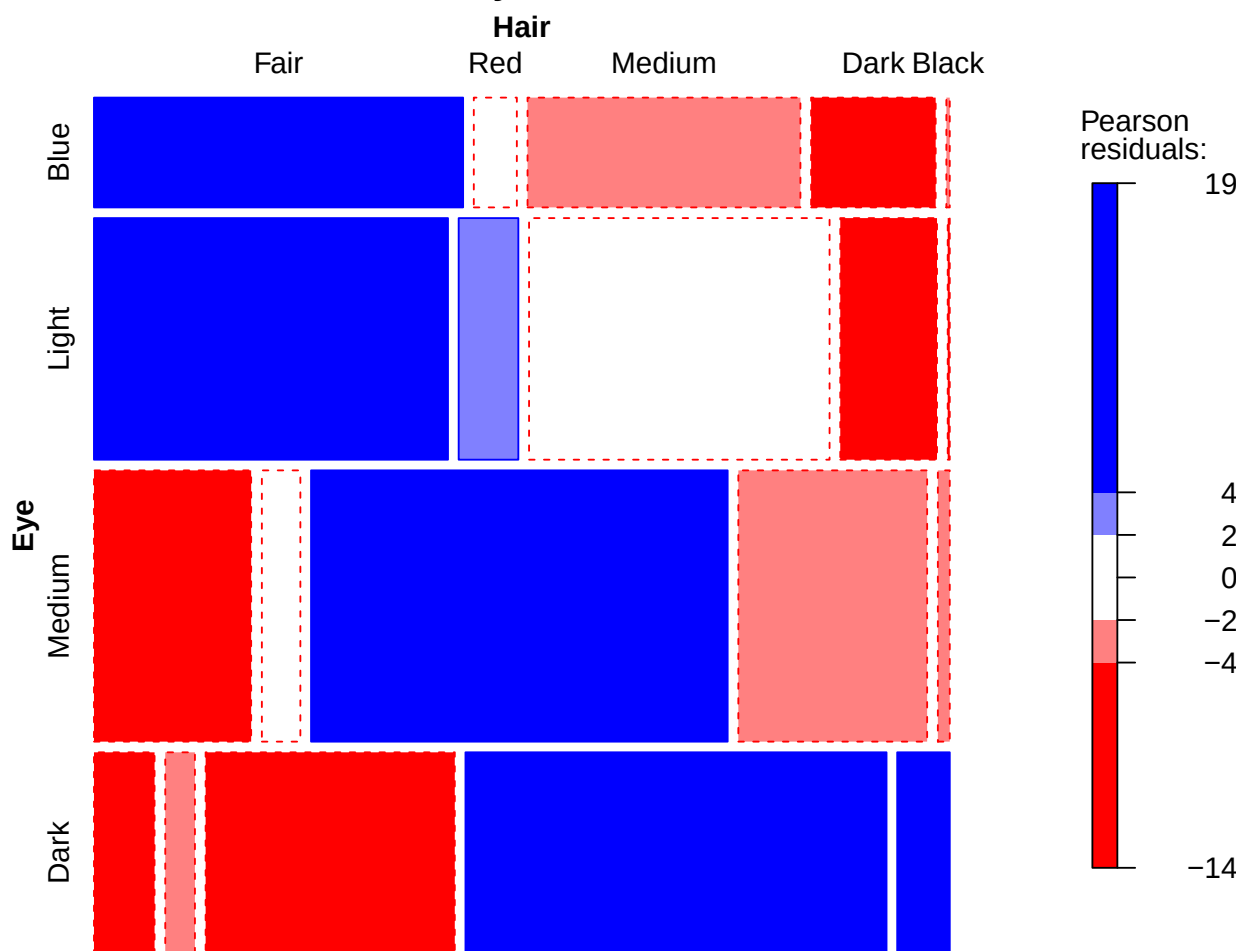
Part (a)

Prepare separate mosaic displays, one for each of Caithness and Aberdeen. Comment on any difference in the pattern of residuals.

```
data("HairEyePlace", package = "vcdExtra")

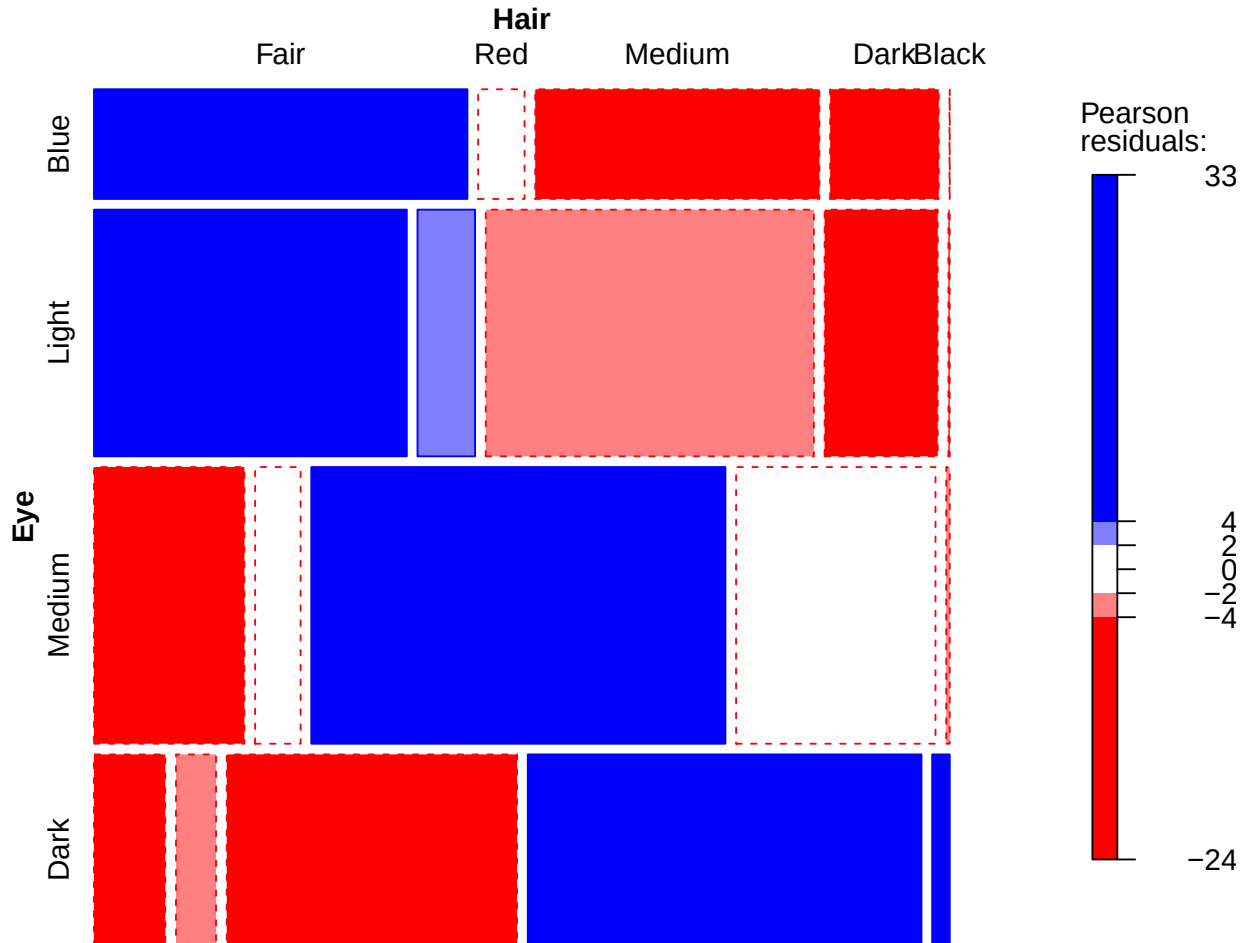
mosaic(HairEyePlace[,,"Caithness"], shade = TRUE, gp = shading_Friendly,
  main = "Hair Colour Versus Eye Colour\nin Caithness From the\nHairEyePlace Dataset")
```

Hair Colour Versus Eye Colour in Caithness From the HairEyePlace Dataset



```
mosaic(HairEyePlace[,,"Aberdeen"], shade = TRUE, gp = shading_Friendly,
  main = "Hair Colour Versus Eye Colour\nin Aberdeen From the\nHairEyePlace Dataset")
```


Hair Colour Versus Eye Colour in Aberdeen From the HairEyePlace Dataset



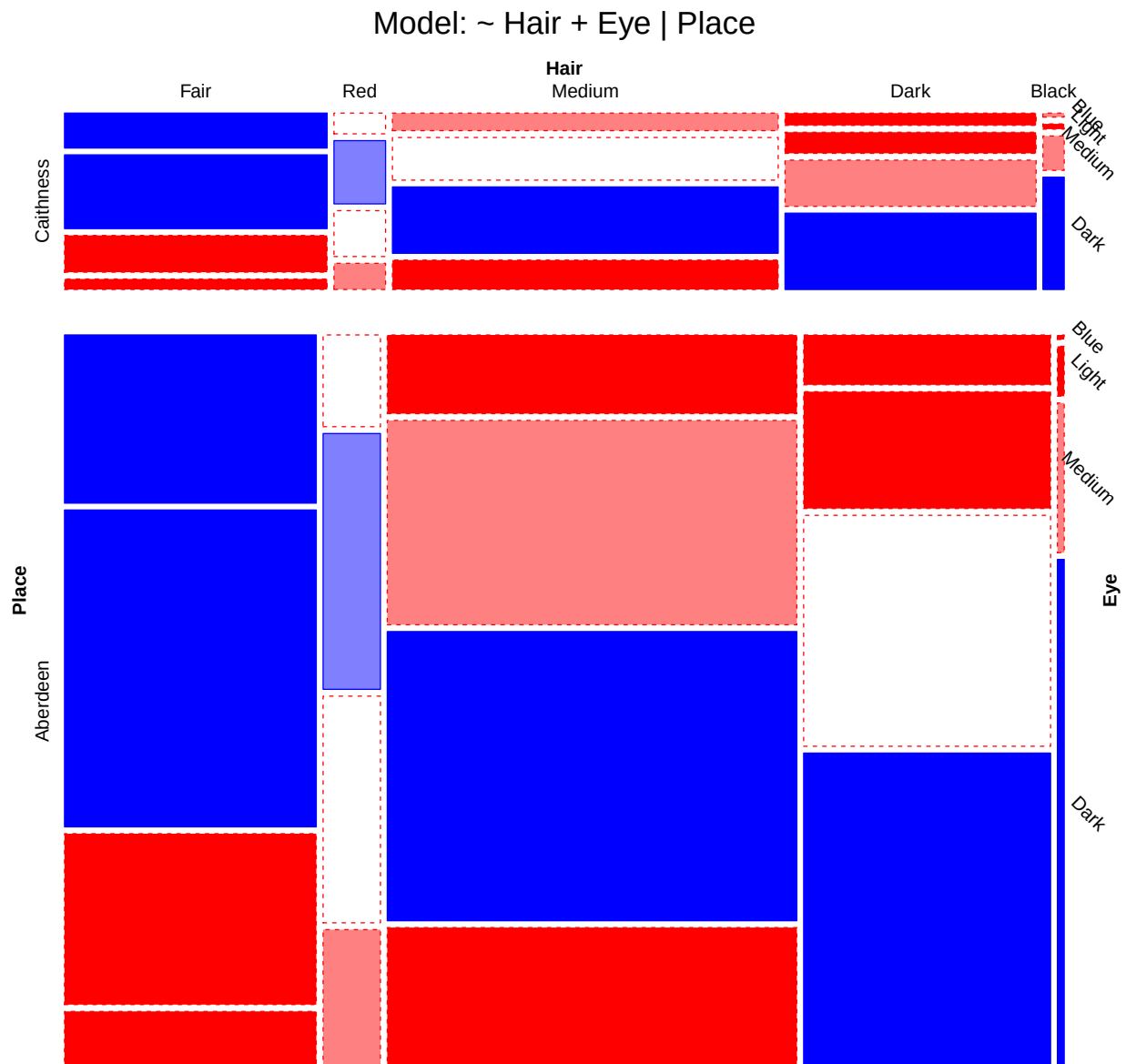
The pattern of residuals is largely the same, with brightness of eyes/hair being associated with larger than expected frequencies of a similar brightness of eyes/hair (compared to a model of independence). Thus, in both displays there appears to be a pattern of agreement where diagonal elements have large positive residuals, and larger negative residuals tend to be further from the diagonal.

On average, data from Aberdeen exhibit a larger magnitude of residuals than the data from Caithness. The pattern of agreement/disagreement is more evident in the Aberdeen data due to the cells corresponding to medium hair reflecting larger negative residuals (in magnitude).

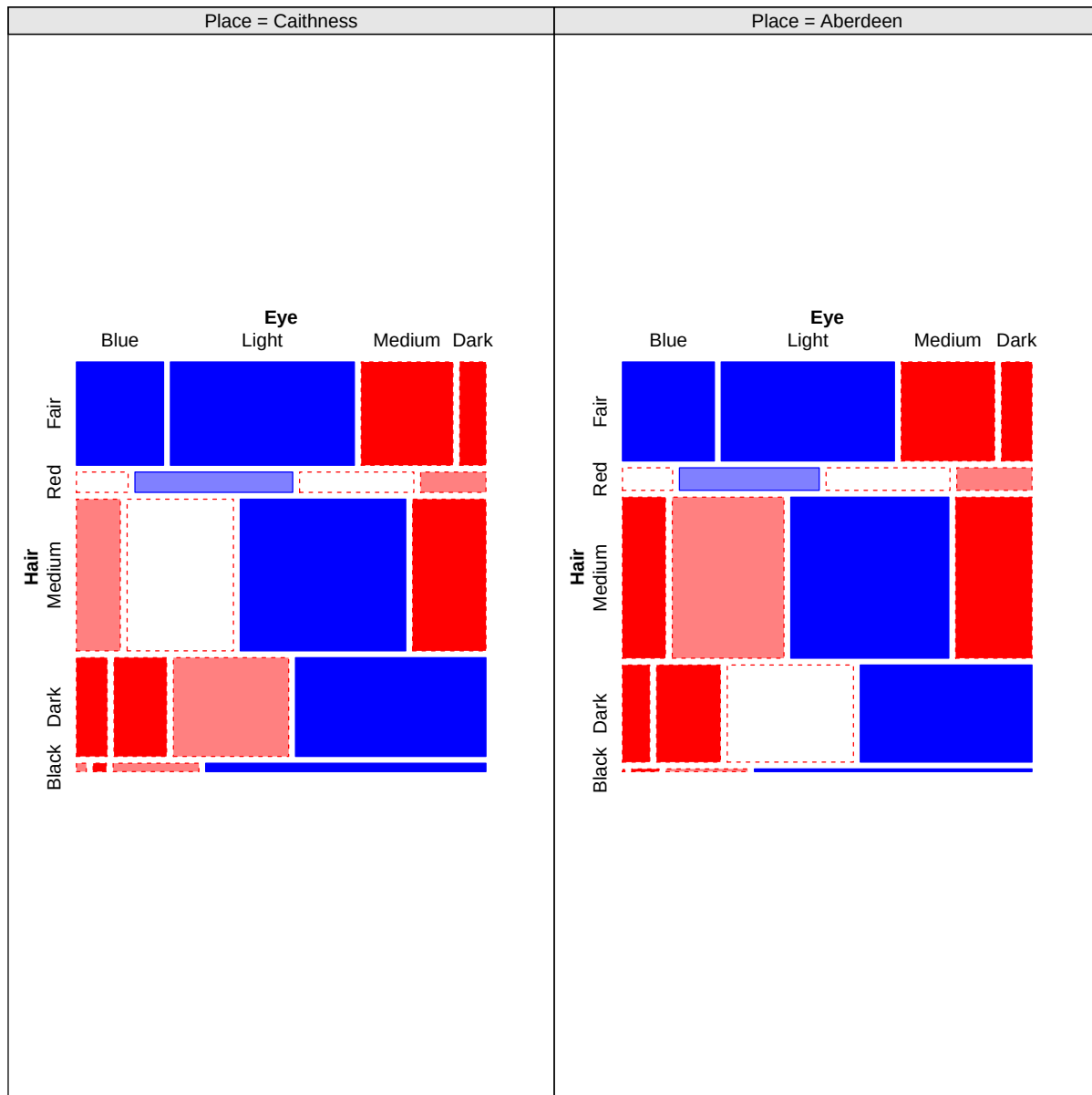
Part (b)

Construct conditional mosaic plots, using the formula $\sim \text{Hair} + \text{Eye} \mid \text{Place}$ and both `mosaic()` and `cotabplot()`. It is probably more useful here to suppress the legend in these plots. Comment on the difference in what is shown in the two displays.

```
mosaic(  
  ~Hair + Eye | Place, data = HairEyePlace,  
  shade = TRUE, gp = shading_Friendly, legend = FALSE,  
  rot_labels = c(right = -45),  
  main = "Model: ~ Hair + Eye | Place"  
)
```



```
cotabplot(
  ~Hair + Eye | Place, data = HairEyePlace,
  shade = TRUE, gp = shading_Friendly, legend = FALSE
)
```



The mosaic display allows for a comparison of relative sample size between Caithness and Aberdeen, whereas the cotabplot allows for easier inspection of residuals within the place (Caithness) that has a lower relative sample size.

Appendix

gavinTheme()

```
gavinTheme <- function(){  
  #' gavinTheme  
  #'  
  #' Theme for ggplot2 graphs.  
  #'  
  #' Used by adding to existing ggplot2 object.  
  #' @examples  
  #' ggplot(...) + gavinTheme()  
  
  theme_minimal() +  
  theme(  
    plot.title = element_text(hjust = .5, margin = margin(r = 20, b = 10)),  
    axis.ticks = element_line(colour = "black", size = .5),  
    axis.line = element_line(colour = "black", size = .5),  
    axis.text = element_text(size = 10),  
    axis.title.x = element_text(margin = margin(t = 10)),  
    axis.title.y = element_text(margin = margin(r = 10)),  
    panel.grid.major = element_blank(),  
    panel.grid.minor = element_blank(),  
    plot.margin = margin(t = 10, r = 20, b = 10, l = 10)  
  )  
}
```